Techfest 2026-27

The Wrecking Ball Challenge

Objective:

To design, build, and deploy an autonomous, battery-powered spherical robot that navigates a 75-meter obstacle-laden arena using vision-based navigation, center-of-gravity stabilization, and real-time obstacle evasion.

Challenge Question:

How can student engineering teams design a self-contained spherical robot that balances internal mechanical movement with real-time vision algorithms to autonomously navigate ramps, smash low-density barriers, and time its movement to evade a swinging pendulum over a 75-meter track?

Scope & Required Deliverables

Hardware & Mechanical Constraints

- **Robot Structure:** Must be a spherical robot (ball-shaped) with a strict diameter between **2.0 feet and 2.5 feet**.
- **Internal Housing:** All core computing hardware, power sources, actuators, and electronics must fit completely inside the spherical body.
- **Peripheral Limits:** External sensors, mounts, or camera lenses cannot extend more than **0.5 feet** beyond the surface of the outer shell.
- **Power Supply:** Must be strictly onboard battery-powered. No tethered cables, external power lines, or remote power feeds are allowed.

Deliverables

1. **Phase 1 CAD & System Document:** Complete CAD assembly illustrating internal electronic layout, center-of-gravity placement, and system architecture schematic.
2. **Phase 2 Technical Simulation Report:** Mathematical analysis validating actuator torque across inclined planes, balance stability, and safety kill-switch mechanism implementation.

3. **Phase 3 Physical Prototype:** Fully operational ball bot fielded live on-campus at IIT Bombay for technical evaluation and track trials.

Track-Specific Environment Stack

The 75-Meter Arena Layout

- **Visual Guidance:** Track boundaries are defined by color-coded cone barriers:
  - **Red Cones:** Located on the right side of the track.
  - **Yellow Cones:** Located on the left side of the track.
- **Sector 1: The Gate Runway (0m - 15m)**
  - Straight acceleration stretch featuring wide chevron-patterned cone gates to allow onboard vision calibration.
- **Sector 2: The Step-Up Ramps (15m - 35m)**
  - Three consecutive incline ramps with heights of **0.2 ft, 0.4 ft, and 0.8 ft**, funneled through cone gates to test traction and motor torque.
- **Sector 3: The Wrecking Zone (35m - 55m)**
  - Sector populated with low-density obstacles (foam, thermocol, cardboard). The robot must maintain course alignment using border cones while plowing through obstacles.
- **Sector 4: The Pendulum Gauntlet (55m - 75m)**
  - Final hazard sector featuring a continuously swinging mechanical pendulum across the track before the finish line.

Format of Competition:

- **Round 1 (Abstract & Design Audit):** Online screening based on CAD models, system schematics, and dimension compliance.
- **Round 2 (Virtual Simulation Audit):** Virtual evaluation of motor sizing calculations, torque requirements for the 0.8 ft ramp, and wireless kill-switch mechanics.
- **Round 3 (On-Site Finals at IIT Bombay):**
  - **Day 1:** Technical inspection (size check, weight, electrical isolation, emergency kill-switch validation) and preliminary practice runs.
  - **Day 2:** Official timed final heats.

Evaluation & Scoring Criteria

- Each participating team is granted **3 restart attempts** within their designated operational window, with only their top performance logged for official scoring.
- **Scoring Formula:**
  $FinalScore(Seconds) = RawRunTime + Penalties$
- Victory is awarded to the competitor achieving the lowest overall **Final Score**.
- For unfinished runs resulting in a DNF status, standings are evaluated solely on total displacement traversed along the 75-meter track.

Team Specification & Eligibility

- **Team Size:** Maximum of 6 members per team.
- **Eligibility:** Open to all currently enrolled undergraduate, postgraduate, or diploma students from recognized academic institutions. Inter-college teams are permitted.

Timeline & Key Dates

Phase	Event / Milestone	Deadline / Schedule
Phase 1	Registration & CAD Abstract Submission	October 25, 2026
Phase 2	Virtual Design & Simulation Audit Submission	November 15, 2026
Phase 2 Result	Finalist Announcement	November 20, 2026
Phase 3 (Day 1)	On-Site Inspection & Qualifying Heats (IIT Bombay)	December 16, 2026
Phase 3 (Day 2)	Grand Finale Runs (IIT Bombay)	December 17, 2026

General Rules of the Competition

1. All teams must register on techfest.org to receive an official Team ID for tracking and submissions.

2. Instructions for specific rounds would be communicated on time via registered mail IDs
3. Submissions must represent the team's original work. Using external AI tools during live rounds is strictly prohibited unless judges explicitly authorize it.
4. Official announcements will be posted on techfest.org. All contest queries must be submitted through the official portal's Doubt section.
5. Decisions made by the judges and organizing committee are final and binding on all participants.
6. **Autonomy:** The robot must operate with 100% autonomy during its run. No manual remote control, external human intervention, or external processing offloads are allowed once the run begins.
7. **Sensor Restrictions:** Onboard cameras must serve as the primary sensor for path tracking and gate identification. Supplementary internal sensors (IMUs, encoders, distance sensors) are permitted for stabilization.
8. **Cone Contact Penalties:** Touching, knocking over, or running over any track cone incurs a **+5 second penalty** per instance added to the total run time.
9. **Emergency Stop:** Every robot must have an operational wireless emergency kill switch that can immediately stop all motor actuation during a run.
10. **Did Not Finish (DNF):** A run is marked as DNF if the robot fails to move, loses complete control, suffers total power failure, or cannot clear an obstacle.

Certificate Policy

- **Certificate of Merit:** Awarded to top-performing podium teams.
- **Certificate of Participation:** Issued to all teams that clear the Phase 1 submission and present their physical prototype for technical inspection at the IIT Bombay Grand Finale.

Prize Money

Position	Rewards
Winner (Rank 1)	**₹2,50,000 Cash** + 3-Month Paid Engineering Internship at Ultraviolette Automotive + 3-Month Claude Pro Subscription + Techfest Merit Certificate
Runner-Up (Rank 2)	**₹1,00,000 Cash** + Techfest Merit Certificate

Terms & Conditions for Prize Money:

- **Track Completion:** Prize money will be awarded **only** if the respective teams successfully finish the track.
- **Disbursement:** The prize money will be directly disbursed to the winning teams by the Ultraviolette team. Techfest holds no responsibility or liability for the distribution or processing of the cash prizes.